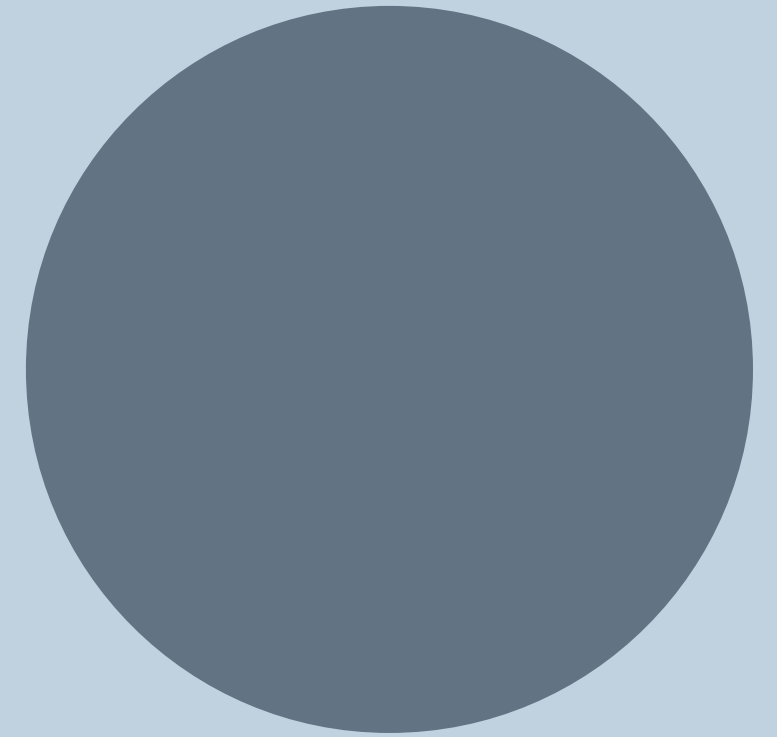
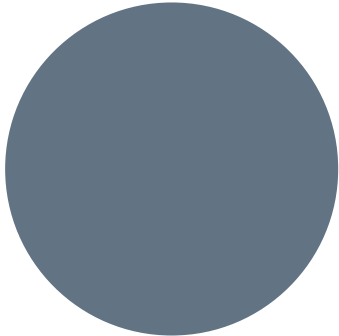




From Rainfall to Risk: Predicting Flood Hazards

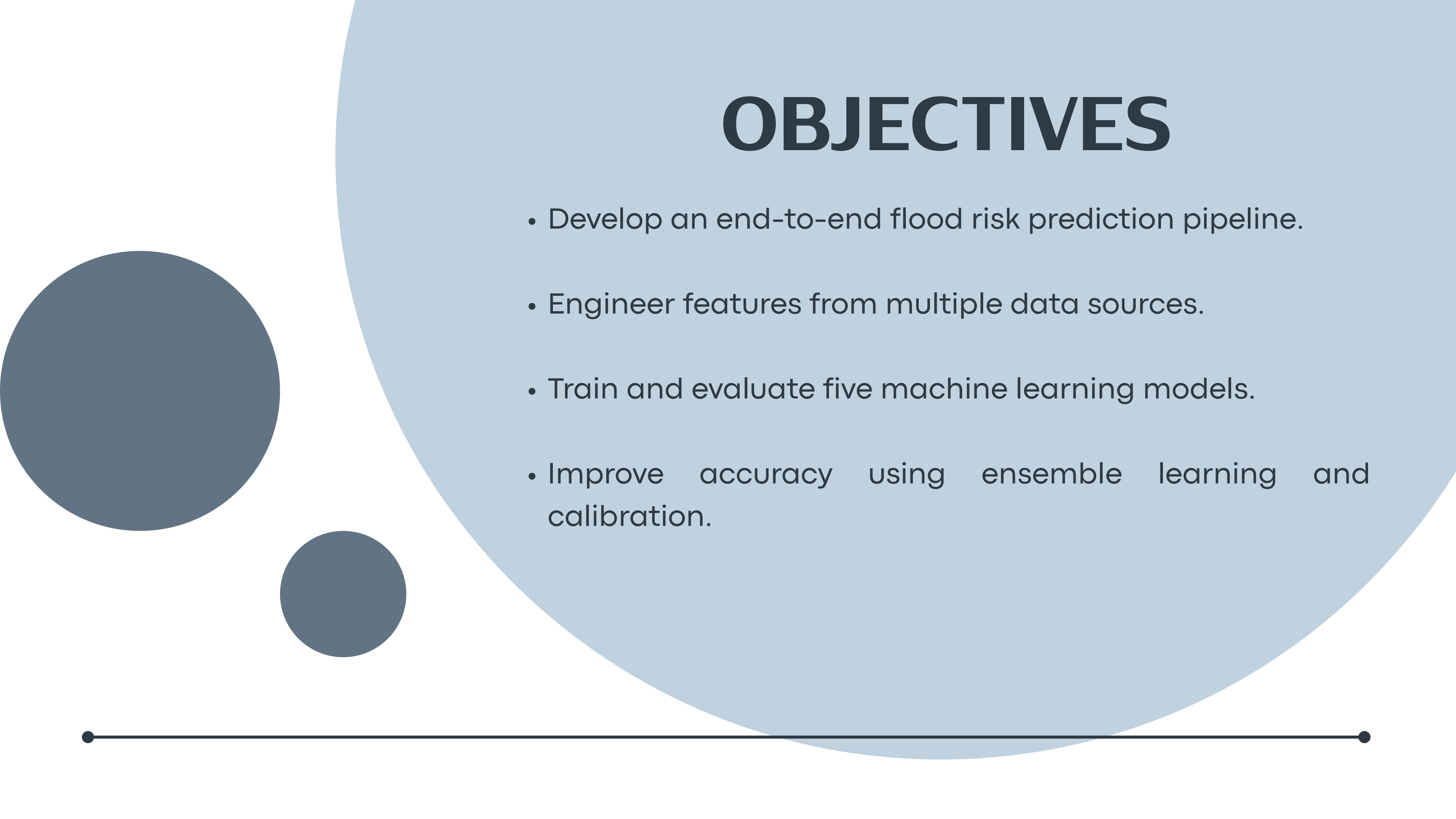
The Outliers_MLOps



INTRODUCTION

- Flooding is one of the most damaging natural disasters in Sri Lanka, driven by monsoonal rainfall, river overflows, and rapid urbanisation
- Accurate, location-specific flood risk prediction is essential for effective disaster preparedness, infrastructure planning, and resource allocation
- This project models flood risk as a continuous score (0–1) rather than a simple category, allowing finer distinction between areas
- For practical decision-making, the score is also translated into four risk levels: Low, Moderate, High, and Very High





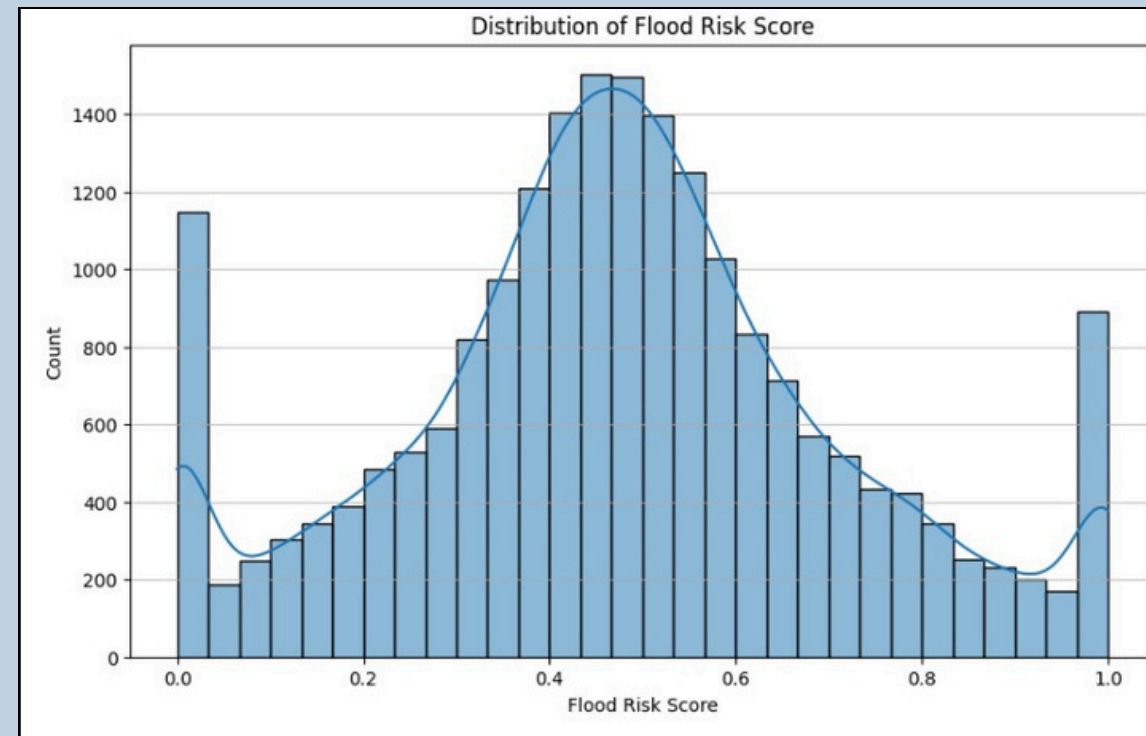
OBJECTIVES

- Develop an end-to-end flood risk prediction pipeline.
- Engineer features from multiple data sources.
- Train and evaluate five machine learning models.
- Improve accuracy using ensemble learning and calibration.

METHODOLOGY

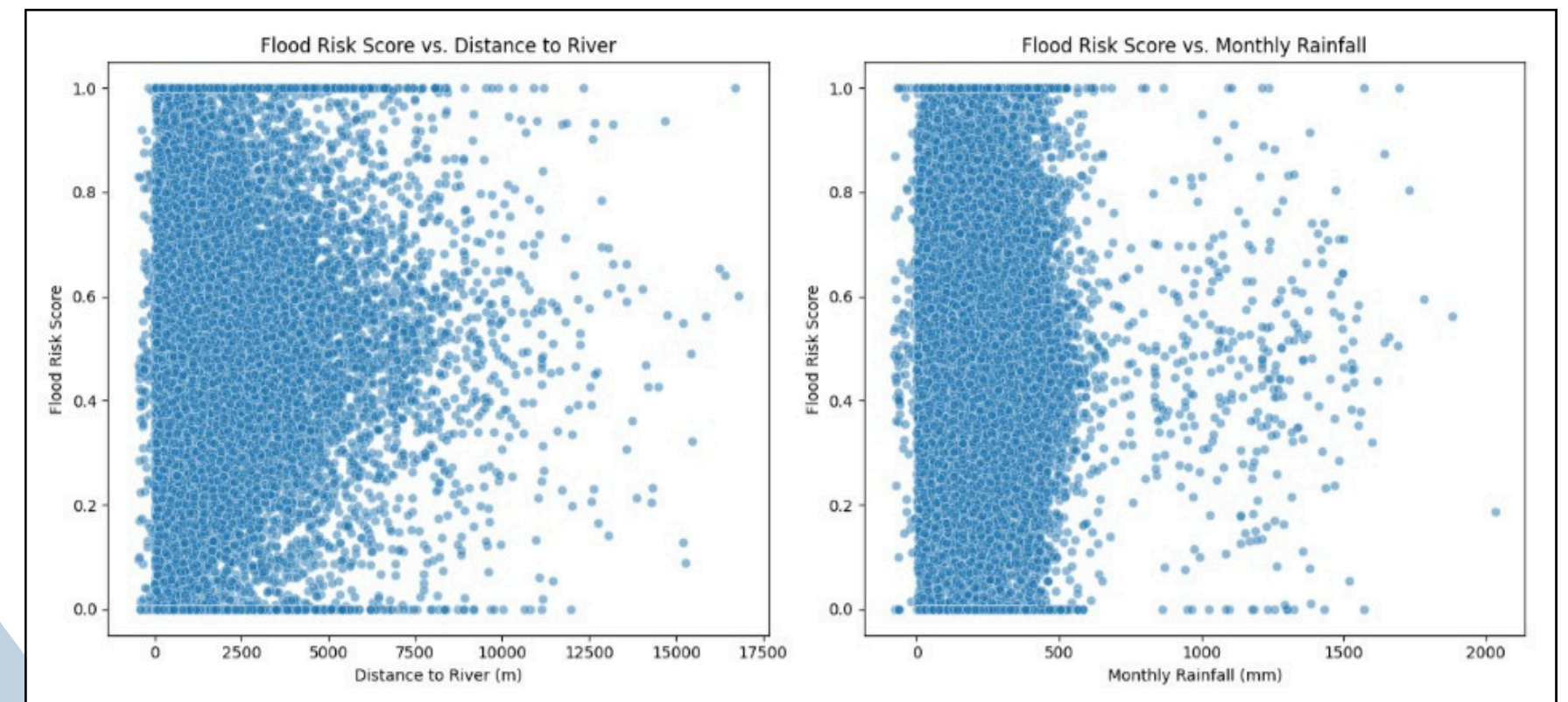
1. Data Cleaning & Missing Value Handling
2. Feature Engineering
3. Feature Selection (Top 150 Features)
4. Train 5 Machine Learning Models
5. 10-Fold Cross-Validation
6. Ensemble Learning (Blending & Stacking)
7. Isotonic Calibration & RMSE Evaluation

DATA ANALYSIS

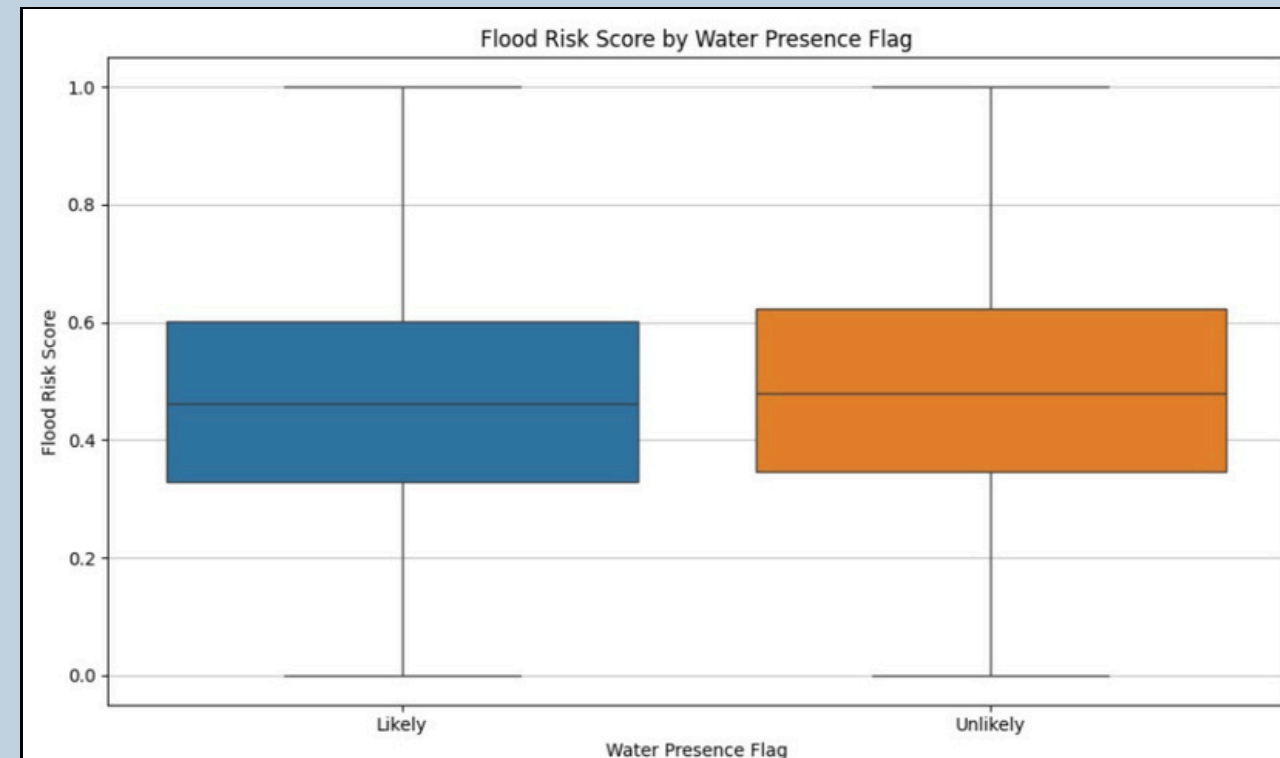


The Flood Risk Score centers normally around 0.45 to 0.50 but shows heavy, anomalous spikes at exactly 0.0 and 1.0. This boundary accumulation indicates the data is censored or inflated, meaning standard regression will be biased at the edges.

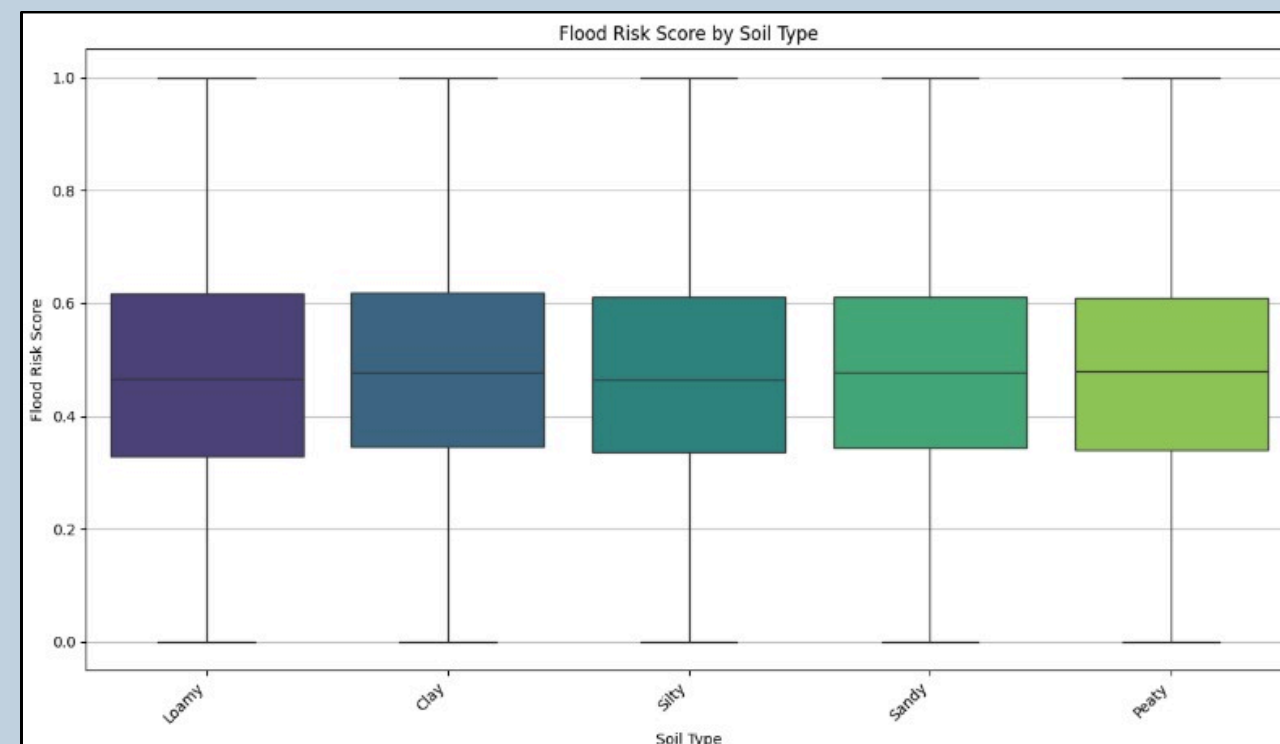
The scatter plots evaluate Flood Risk Score against Distance to River and Monthly Rainfall. Both variables show a high density of data points concentrated at lower ranges (under 5,000 meters for river distance and under 500 mm for monthly rainfall) before thinning out at higher values. Crucially, both plots show heavy horizontal lines of data points locked at the absolute boundaries of 0.0 and 1.0 across all levels of the predictors.



DATA ANALYSIS



The box plot shows that Flood Risk Score is nearly identical whether the Water Presence Flag is "Likely" or "Unlikely." Both groups share almost the same median score (around 0.47) and the exact same spread from 0.0 to 1.0. Because the two boxes overlap almost completely, this flag by itself does not help separate or predict different levels of flood risk.



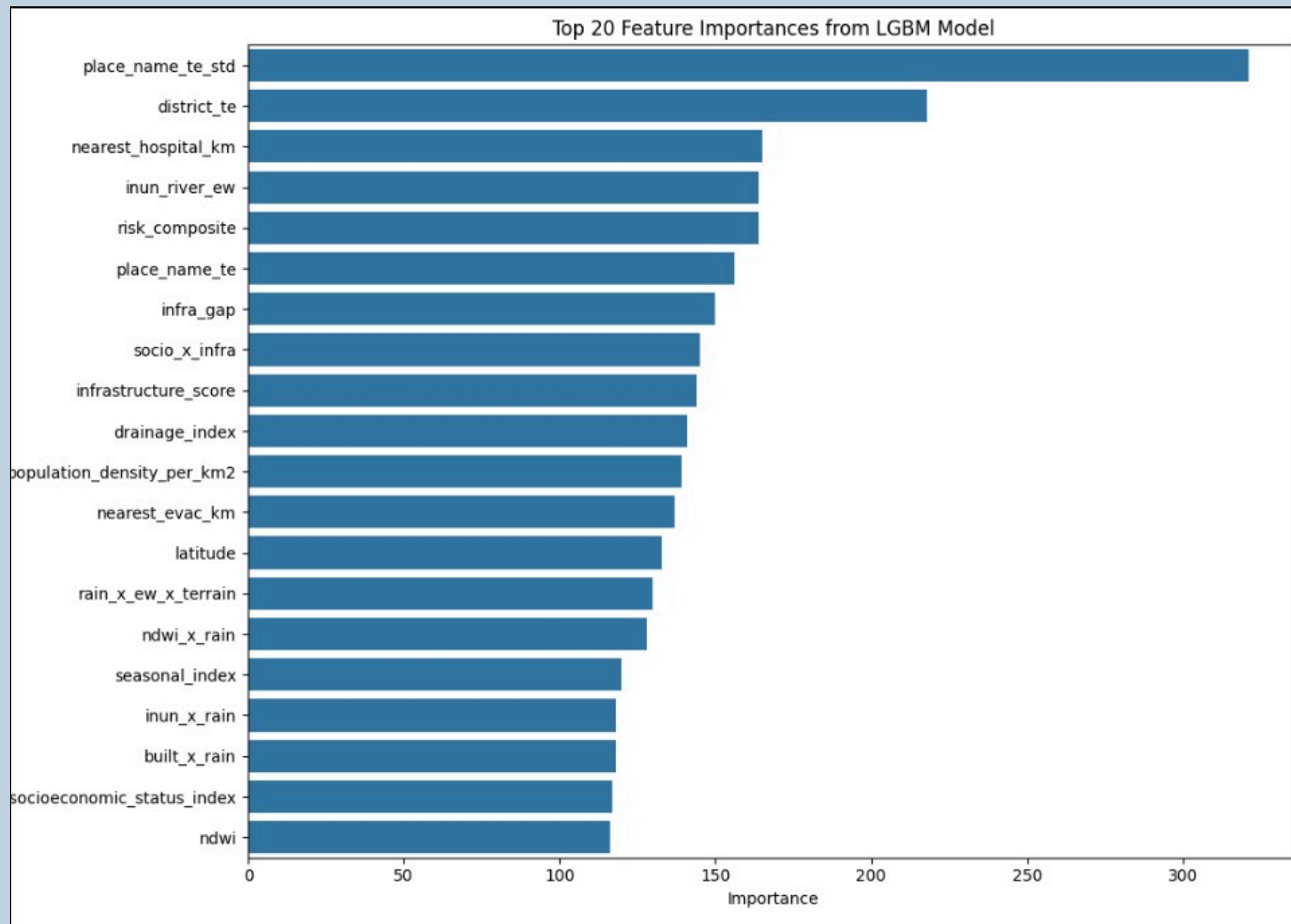
The box plot shows the Flood Risk Score by Soil Type and it indicates that the Flood Risk Score is nearly identical among all soil types. All groups share almost the same median indicating that the soil types themselves do not predict different levels of flood risk.

DATA ANALYSIS

Metric	Value	Interpretation
OOF RMSE (post-calibration)	0.23428	Average prediction error of ~0.234 risk units on a 0–1 scale
R-squared (R^2)	0.03721	Model explains ~3.7% of variance in flood
Calibration Method	Isotonic Regression	Non-parametric monotone calibration on OOF predictions
Evaluation Strategy	Out-Of-Fold (OOF)	Unbiased generalization estimate via K-Fold
Metric Type	Regression	Continuous target; RMSE and R^2 used

- **Baseline Accuracy:** Achieved an Out-of-Fold (OOF) RMSE of 0.234 on a normalized $[0, 1]$ risk scale.
- **Operational Readiness:** A ~23% error margin is highly effective for regional prioritization and resource allocation.
- **Low R^2 (0.037):** Expected and consistent with complex, real-world geospatial regression.

FEATURE IMPORTANCE



- Location is the strongest predictor
- Infrastructure and social vulnerability matter
- Engineered hydrological features work better than raw data
- Healthcare access shows mixed effects

DISCUSSION:

MODEL LIMITATIONS

- Low explanatory power ($R^2 = 0.037$): Large portion of flood risk variance remains unexplained, common in geospatial risk modeling.
- Uncertain labels: Flood risk estimates based on historical/simulated data contain noise, especially for rare events.
- Spatial dependence: Standard cross-validation may underestimate error due to spatial autocorrelation.
- Limited temporal dynamics: Time-varying factors (e.g., antecedent rainfall, seasonal effects) are not fully captured.
- Data gaps: Sparse infrastructure and socioeconomic data may bias results toward urban areas.



RECOMMENDATIONS FOR IMPROVEMENT

- Spatial cross-validation: Use spatial block CV for more reliable performance estimates.
- Richer hydrological features: Add terrain and flow-based variables (e.g., slope, catchment area).
- Stronger ensembles: Incorporate models like XGBoost and CatBoost with stacking/blending.
- Temporal feature engineering: Include rolling rainfall and moisture indicators.
- Hybrid modeling: Integrate outputs from physics-based models (e.g., SWAT, HEC-RAS).
- Active learning: Target uncertain predictions for improved data collection.

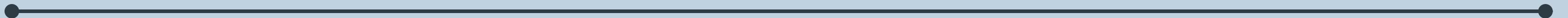
TECHNIQUES USED

- Programming language: Python 3.9+
- Core ML Framework: LightGBM
- Data Processing: Pandas, Numpy
- Feature Engineering: Scikit-learn
- Calibration: Sklearn.isotonic
- Cross-Validation: Sklearn.model_selection
- Evaluation: Sklearn.metrics
- Visualization: Matplotlib, Seaborn
- Experiment Tracking: MLflow/ Custom logger



CONCLUSION

- Developed a production-ready flood risk prediction system using a robust MLOps pipeline.
- A LightGBM model with calibration achieved an RMSE of 0.23, providing a reliable baseline for continuous flood risk scoring at district and sub-district levels.
- Key predictive factors include geographic location, infrastructure quality, socioeconomic vulnerability, and hydrological interactions—consistent with existing research.
- Although overall variance explained is low ($R^2 = 0.037$), model accuracy is sufficient for risk prioritization and resource allocation.
- Future work will focus on spatial validation, richer ensembles, temporal weather features, and integration with physics-based flood models.



THANK YOU
